# Sanskrati Agarwal

Senior Undergraduate
Department of Mechanical Engineering
Sanskrati Agarwal | Sanskrati12
sanskrati23 | Portfolio

| Year | Qualification | School/Institution | CPI/% |
|---|---|---|---|
| 2023 – Present | B.Tech | Indian Institute of Technology Kanpur | 7.1/10.0 |
| 2023 | Class XII (CBSE) | Dr Lokmandas Public School, Etah | 95.8% |
| 2021 | Class X (CBSE) | Assisi Convent School, Etah | 98.8% |

## WORK EXPERIENCE

### Investywise by Groww — *Jun'26 - Jul'26*

*Full Stack Developer Intern (Remote)*

- Enhanced a **real-time data pipeline (Python, REST APIs, n8n)** polling **BSE filings** every **30 seconds** and converting them into published articles within **10 seconds**
- Diagnosed and resolved a **production-breaking duplicate content bug** via **root cause analysis**, implementing **fuzzy matching (75% similarity threshold)** and extending the dedup window **6** times (**5→30 min**), cutting duplicate detection latency to under **15ms**
- Uncovered and fixed a critical **data pagination bug** that was silently dropping dropping **50 %** of fillings every cycle restoring **full content coverage**
- Automated **SEO** and social metadata generation using **Gemini AI** and the **WordPress REST API**, boosting discoverability
- Independently managed production infrastructure, **AWS EC2, Docker, CI/CD (Gitea)** including deployments & monitoring

## KEY COMPETITIONS

### ML Hackathon (IITKGP Fugacity 2026) — *Aug'26*

*Ranked in top **30** among **374 teams**, qualifying for **final round***

- Developed a physics informed **hybrid ML model** to predict **Product B yield** using only **150** given labelled samples
- Engineered reaction aware features including **residence time, effective temperature, temperature differential** and **Arrhenius kinetics** to capture non-linear yield behaviour
- Combined a **5-parameter** kinetic model with residual **XGBoost**, achieving **13.38 RMSE** via leakage-safe, **25-fold repeated cross-validation**
- Evaluated **8** ML models, predicted **50** unseen reactor conditions, enforcing physical constraints and tuning **hyperparameters**

### Accenture Innovation Challenge 2026 — *Aug'26 - Sep'26*

*BusinessIntelligence.ai*

- Built an AI powered **KPI** storytelling engine to detect **business anomalies**, perform **Root Cause Analysis** and generate **actionable recommendations**
- Integrated **structured KPIs** and **unstructured signals** to generate **evidence-backed** insights across multiple data sources
- Proposed a **correlation to causation** framework using temporal and comparative evidence with **confidence scored hypotheses**
- Designed an **AI RCA** & **CAPA engine** to convert multi-source business data into traceable and actionable recommendations

## RELEVANT COURSES

| | |
|---|---|
| Fundamentals of Computing I & II | Linear Algebra |
| Introduction to Electronics | Complex Variables |
| Partial Differential Equations | Multi Variable Calculus |
| Machine Learning for Engineers | Critical Thinking |

## EXTRA CURRICULARS

- Secured **1st** position in **Formula Bharat 2026 Rulebook Quiz** in **EV category** among **40+** FSAE Indian teams
- Qualified for **Stage-II** in **National Talent Search Examination** (NTSE) among **10 lakh+** candidates
- Secured **State Rank 3** in **UP Genius 2017**, youngest finalist from Class **6**, competing with students up to **12th**
- Secured **Gold Medal** in **Chess** at **Inferno**, the **Inter-Hall Competition** among **30+** participants from different batches
- Secured **3rd Rank** in UP among **1000+** candidates in the Inter-State **Anuvrat Essay Writing Competition**

## KEY PROJECTS

### SenseAI - AI Career Coach Platform

*Self Project* — *Jun '26 - Jul '26*

- Developed an **AI career coaching platform** using **Next.js, Prisma, PostgreSQL, Clerk** delivering resume building and analysis, mock interviews, and personalized career guidance
- Engineered an **AI resume analyzer** with **Google Gemini** to evaluate **ATS** score and targeted improvement suggestions
- Designed a normalized **Prisma/PostgreSQL schema** for users, resumes, cover letters, and interview analytics, with an **interactive dashboard** visualizing performance trends
- Implemented **PDF export** feature for AI-generated resumes

### WhatToWatch - AI Enhanced Movie Platform

*Self Project* — *May '26 - Jul '26*

- Built a **movie and TV discovery platform** using **React, Node.js, Express, and MongoDB**, integrating the **TMDB API** for dynamic content and deployed on **Render**
- Engineered a **Gemini-powered recommendation engine** that analyzes 5 mood/taste-based inputs to generate personalized movie suggestions beyond conventional genre filtering
- Developed secure **REST APIs** with **JWT authentication (bcrypt)** for user accounts, movie data, and interactions, maintaining a decoupled frontend-backend architecture
- Built dynamic movie/TV pages with ratings, revenue, budget, and **YouTube trailer integration**, using **Zustand** for state management and **Axios** for API communication

### SPLITR - AI Expense Management Platform

*Self Project* — *Apr '26 - Jun '26*

- Architected a real time **expense management platform** using **Next.js, React, and Convex**, enabling real-time expense tracking, group splitting, and settlement management
- Designed a **Convex data model** for users, groups, expenses, and settlements, supporting individual and multi-user expense splitting with automated participant-wise balance calculations
- Implemented **Clerk authentication** with webhook-driven user synchronization to Convex, ensuring consistent user records across the application
- Built scheduled **Inngest workflows** delivering Gemini-generated **spending insights** and **payment reminders** via **Resend**, with a responsive UI (Tailwind CSS, Shadcn UI) deployed on **Vercel**

### Drone Flight Analysis

*Course Project, Prof Malay Das* — *Feb '26 - Apr '26*

- Built an end-to-end ML pipeline to classify **drone flight stability** and **battery consumption** from **544,000+** IMU rows
- Trained **Logistic Regression** and **Random Forest**, achieving **96.15%** & **79.41%** accuracies for **stable flight** detection
- Analyzed **PX4 flight logs** with **PyULog** and estimated remaining flight and predicted current consumption

## TECHNICAL SKILLS

- **Languages & Tools:** C++, Python, Javascript, Typescript, SQL, MATLAB, Git, Docker, AWS, Prisma, Convex
- **Libraries :** NumPy, Pandas, Matplotlib, Scikit-learn, Streamlit

## POSITIONS OF RESPONSIBILITY

### Secretary, Chess Club (IIT Kanpur) — *Jun'24 - May'25*

- Authored informative **articles, emails, and social media posts** promoting chess events across the campus community
- Led registration for 80+ teams for the **India Collegiate Chess Championship Fall 2024**, hosted by **Chess.com**
- Coordinated **Chess Master Premier League** (CMPL) which attracted **5000+** participants from **80+** colleges across India